

An Affordable Recipe for Training Diffusion Transformers: Dog-Breed Text-to-Image Diffusion, from Pretraining to Fine-Tuning, on 4 Consumer GPUs

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Training diffusion transformers is usually presented as something that requires industrial-scale compute. This resource teaches the opposite lesson through a real, runnable pipeline arrived at over two months of empirically comparing image tokenizers, latent-space parameterizations, and training hyperparameters, converging on a configuration that pretrains a working text-to-image Diffusion Transformer (DiT) to sub-10 FID in 2.5 hours, and fine-tunes it in 20 more minutes, on hardware any student can rent by the hour. Students run this recipe themselves and evaluate their own results — no pretrained weights are distributed, so every checkpoint you get is your own.

~2h50m total training on 4×RTX 3090 (24GB) → sub-10 FID — about \$3–\$5 of rented compute for the whole pipeline.

2.5h pretrain (Exp. 1) + 20min fine-tune (Exp. 2). GPU pricing: Vast.ai \$0.25/GPU/hr, RunPod \$0.46/GPU/hr (Aug 2026 quoted rates).

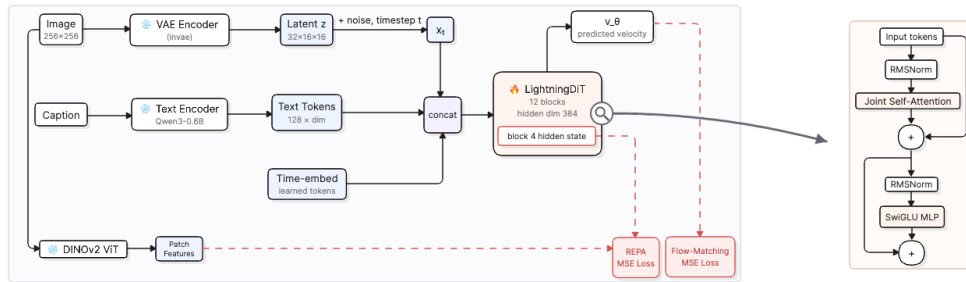


Figure 1: Training architecture, at a glance. Frozen components (snowflake: VAE encoder, text encoder, DINOv2 target) supply latents, text tokens, and REPA targets; only LightningDiT (fire) is trained, jointly on the flow-matching velocity loss and the REPA feature-alignment loss tapped at block 4. Right: internals of one of the 12 identical pre-norm transformer blocks (RMSNorm → joint self-attention → residual → RMSNorm → SwiGLU MLP → residual).

Learning objectives.

- Explain latent-space diffusion and derive the flow-matching training objective (straight-line paths, velocity prediction).
- Contrast in-context (token-based) conditioning against AdaLN-Zero, and apply REPA-style auxiliary losses that align internal DiT features with a pretrained encoder.
- See how tokenizer/recipe choices, not scale, determine whether DiT training is affordable on consumer GPUs, and build a full data → precompute pipeline.
- Diagnose what a fine-tuning FID curve really measures, and score generations with learned human-preference models (PickScore, HPSv2) against FID/IS.

What students actually do. The full notebook (13 sections) ties background theory to the real training code, runs each model component individually (VAE tokenizer, text encoder, REPA target, the DiT backbone), builds the data pipeline, runs pretraining and fine-tuning, and evaluates with FID/IS and learned human-

preference models. Every code cell is real, executable code from the actual research codebase, not a simplified reimplementation.

Results below are from my own runs, shown as a concrete example of the phenomena students are asked to reproduce and diagnose themselves in Sections 10 and 12 of the notebook.

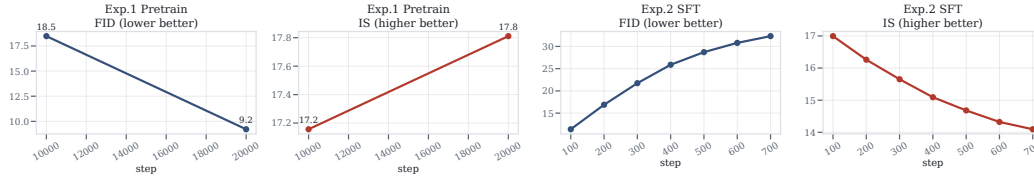


Figure 2: **Exp. 1, pretraining** (vae-repa-recaptioned-latents-invae-EMA0.9995-200Epochs): FID improves 18.5→9.2 and IS improves 17.2→17.8 as pretraining progresses — healthy convergence. **Exp. 2, fine-tuning** (invae-sft-dogs-synthetic-2k-EMA0.995-100ep-5): FID rises 11.4→32.3 and IS falls 17.0→14.1 across the run — see the discussion below for why this is not simply “the model got worse.”

Exp. 3 — learned human-preference scoring (PickScore, HPSv2).

	HPSv2 mean score	PickScore: preferred over other model
Pretrained	0.2409	2.4% of 500 pairs
SFT	0.2719	97.6% of 500 pairs

500 held-out captions, one image per checkpoint per caption, same seed. Both reward models — trained on real human preference judgments, independent of FID/IS — agree the SFT checkpoint is preferred.

Why does FID rise rather than fall? Most likely a metric artifact, not overfitting: both evaluation stages reuse the identical FID reference statistics, built once from the 26,000-image pretraining photo set — including for the SFT model, whose actual training data (2,000 curated synthetic images) is a deliberately different distribution. FID measures distance to whichever reference it’s given, so specializing toward a new distribution raises FID against the old one regardless of quality. The checkpoint scored in Exp. 3 above (ep-0000100, the final and single worst-FID checkpoint) is still the one PickScore/HPSv2 prefer 97.6% of the time — hard to square with “the images got worse,” easy to square with a mismatched reference. Neither metric alone tells the complete story here; students do not see these results before running the notebook, and reproduce and reconcile all three themselves on their own trained checkpoints (Sections 10 and 12).

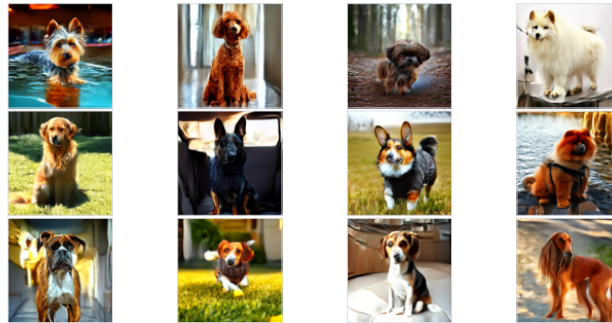


Figure 3: 12 sample generations from the SFT checkpoint (invae-sft-dogs-synthetic-2k-EMA0.995-100ep-5/ep-0000100, the same checkpoint scored in Exp. 3), drawn from the 500-caption reward-model evaluation set.

Resource access. Code, configs, and the full tutorial notebook (dog_t2i_diffusion_tutorial.ipynb): <https://github.com/Reyhaneesmailizadeh/diffusion-bench>. No pretrained weights are distributed with this resource — students download the raw datasets and train both stages themselves, on GPU compute of their own choosing (the notebook documents the exact commands used at the scale run for this submission).